**Result 1 (assumed subpath inclusion).** [Subpath restrictions, inclusion from named universal assumptions]
Let $n$, $a$ and $m$ be natural numbers. Let $h$ be evidence that $a + m \leq n$. Let $d$ be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Assume subpath sources preserved$(n, a, m, h, d)$ (`hs`). Assume subpath targets preserved$(n, a, m, h, d)$ (`ht`). Then

$$\left(\text{there exists } i : \text{ShapeMap}(Q_{d|_{a,m}}, Q_d),\ (\text{for every } v : \{0, \ldots, m+1-1\},\ i.\text{vertices}(v) = a + v)\right).$$

**Result 2 (subpath sources).** [Subpath restrictions, endpoint preservation]
Let $n$, $a$ and $m$ be natural numbers. Let $h$ be evidence that $a + m \leq n$. Let $d$ be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Then

$$\text{subpath sources preserved}(n, a, m, h, d).$$

**Result 3 (subpath targets).** [Subpath restrictions, endpoint preservation]
Let $n$, $a$ and $m$ be natural numbers. Let $h$ be evidence that $a + m \leq n$. Let $d$ be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Then

$$\text{subpath targets preserved}(n, a, m, h, d).$$

**Result 4 (subpath inclusion exists).** [Subpath restrictions, unconditional finite subpath inclusion]
Let $n$, $a$ and $m$ be natural numbers. Let $h$ be evidence that $a + m \leq n$. Let $d$ be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Then

$$\left(\text{there exists } i : \text{ShapeMap}(Q_{d|_{a,m}}, Q_d),\ (\text{for every } v : \{0, \ldots, m+1-1\},\ i.\text{vertices}(v) = a + v)\right).$$

**Result 5 (restricted zigzag object).** [Subpath restrictions, restriction objects]
Let $K$ be a type. Let *field* be a field structure on $K$. Let $n$, $a$ and $m$ be natural numbers. Let $h$ be evidence that $a + m \leq n$. Let $d$ be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Assume subpath sources preserved$(n, a, m, h, d)$ (`hs`). Assume subpath targets preserved$(n, a, m, h, d)$ (`ht`). Let $Z$ be a module diagram on $Q_d$. Let $v$ be a vertex in $\{0, \ldots, m+1-1\}$. Then

$$Z|_{[a,a+m]}.\text{object}(v) = Z.\text{object}(a + v).$$

**Result 6 (restricted zigzag arrow).** [Subpath restrictions, restriction arrows with endpoint transports]
Let $K$ be a type. Let *field* be a field structure on $K$. Let $n$, $a$ and $m$ be natural numbers. Let $h$ be evidence that $a + m \leq n$. Let $d$ be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Assume subpath sources preserved$(n, a, m, h, d)$ (`hs`). Assume subpath targets preserved$(n, a, m, h, d)$ (`ht`). Let $Z$ be a module diagram on $Q_d$. Let $e$ be a vertex in $\{0, \ldots, m-1\}$. Let $Q = Q_{d|_{a,m}}$. Let $R = Q_d$. Let $i = \iota_{a,m}$. Then

$$Z|_{[a,a+m]}.\text{arrow}(e) = (i^* Z)(e).$$

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. This study checks finite zigzag restrictions, including the transported arrow formula; it does not prove the pyramid theorem.